Status of and Threat to Coral Reefs

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Why are coral reefs threatened?

The majority of reef loss or damage is not deliberate. Coral reefs are being degraded by an accumulation of stresses arising from human activities. In simple terms, stresses can be grouped by the actions of people extracting material from, and placing materials upon, coral reefs. Overfishing, pollution and coastal development top the list of chronic stressors. In many situations chronic stresses are overwhelming the resilience, (or the capacity for self-repair), of reef communities. Some coral reefs are covered with sand, rock and concrete to make cheap land and stimulate economic development. Others are dredged or blasted for their limestone or to improve navigational access and safety. In addition to this, long-term changes in the oceans and atmosphere (rising sea temperatures and levels of CO2), and acute stresses from highly variable seasons, severe storms, earthquakes and volcanic eruptions also affect coral reefs.

So what are the different things we are doing to damage coral reefs?

- **Overfishing:** Increasing demand for food fish and tourism curios has resulted in over fishing of not only deep-water commercial fish, but key reef species as well. Overfishing of certain species near coral reefs can easily affect the reef's ecological balance and biodiversity. For example, overfishing of herbivorous fish can also lead to high levels of algal growth. From subsistence level fishing to the live fish trade, inadequate fisheries management is forcing the decline of fish stocks. Choose seafood products that come from certified, well-managed and sustainable fisheries. Certified products are available at most supermarkets - check out the product label, or visit: **www.fishonline.org**
- **Destructive fishing methods:** Fishing with dynamite, cyanide and other methods that break up the fragile coral reef are highly unsustainable. Dynamite and cyanide stun the fish, making them easier to catch. Fishermen say they have no other option if they are to compete with trawlers and overcome a smaller supply of fish because of previous overfishing. These practices generally do not select or target particular fish species and often result in juveniles being killed in the process. Damaging the coral reef habitat on which the fish rely will also reduce the productivity of the area, with further impacts on the livelihoods of fishermen.

- **Unsustainable tourism:** Tourism generates vast amounts of income for host countries. Where unregulated however, tourism pressures can cause damage to the very environment upon which the industry depends. Physical damage to the coral reefs can occur through contact from careless swimmers, divers, and poorly placed boat anchors. Hotels and resorts may also discharge untreated sewage and wastewater into the ocean, polluting the water and encouraging the growth of algae, which competes with corals for space on the reef.
- **Coastal development:** The growth of coastal cities and towns generates a range of threats to nearby coral reefs. Where space is limited, airports and other construction projects may be built on land reclaimed from the sea. Sensitive habitats can be destroyed or disturbed by dredging activities to make deep-water channels or marinas, and through the dumping of waste materials. Where land development alters the natural flow of water, greater amounts of fresh water, nutrients and sediment can reach the reefs causing further degradation. Within the last 20 years, once prolific mangrove forests, which absorb massive amounts of nutrients and sediment from runoff caused by farming and construction, have been destroyed. Nutrient-rich water causes fleshy algae and phytoplankton to thrive in coastal areas in suffocating amounts known as algal blooms. Coral reefs are biological assemblages adapted to waters with low nutrient content, and the addition of nutrients favours species that disrupt the balance of the reef communities.
- **Pollution:** Coral reefs need clean water to thrive. From litter to waste oil, pollution is damaging reefs worldwide. Pollution from human activities inland can damage coral reefs when transported by rivers into coastal waters. Do your bit - do not drop litter or dispose of unwanted items on beaches, in the sea, or near storm drains.
- **Global Aquarium Trade:** It is estimated that nearly 2 million people worldwide keep marine aquariums. The great majority of marine aquaria are stocked with species caught from the wild. This rapidly developing trade is seeing the movement of charismatic fish species across borders. Threats from the trade include the use of cyanide in collection, over-harvesting of target organisms and high levels of mortality associated with poor husbandry practices and insensitive shipping. Some regulation is in place to encourage the use of sustainable collection methods and to raise industry standards.
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How are corals affected by climate change?

- **Coral Bleaching:** Coral bleaching occurs when the symbiosis between corals and their symbiotic zooxanthellae breaks down, resulting in the loss of the symbionts and a rapid whitening of the coral host (thus the term "bleaching"). This is a stress response by the coral host that can be caused by various factors, but more severe and frequent cases are being caused by a

rise in sea surface temperature (SSTs). If the temperature decreases, the stressed coral can recover; if it persists, the affected colony can die.

The impacts from coral bleaching are becoming global in scale, and are increasing in frequency and intensity. Mass coral bleaching generally happens when temperatures around coral reefs exceed 1oC above an area's historical norm for four or more weeks. Sea surface temperature increases have been strongly associated with El Niño weather patterns. However, light intensity, (during doldrums, i.e. flat calm conditions), also plays a critical role in triggering the bleaching response. If temperatures climb to more than 2o C for similar or longer periods, coral mortalities following bleaching increase.

Mass coral bleaching was not documented in the scientific literature before 1979; however, significant mass bleaching events have since been reported in 1982, 1987, 1992 and the strongest sea surface warming event ever recorded occurred in 1998, where an estimated 46% of corals in the western Indian Ocean were heavily impacted or died. In 2005 sea surface temperatures in the Caribbean were the highest reported in more than 100 years, and there was also significant coral bleaching following this warming. This year, coral bleaching is being reported in several locations around the world. If sea surface temperatures continue to rise, then the frequency and severity of coral bleaching will also increase, likely affecting the ability of coral reefs, as we have known them, to adapt and to provide many of the services that people rely upon.

- **Rising sea levels:** Observations since 1961 show that the average temperature of the global ocean has increased even at depths of 3000m (IPCC report), and that the ocean has been absorbing more than 80% of the heat added to the climate system. Such warming causes sea level rise and creates problems for low lying nations and islands.
- **Ocean Acidification:** This is the name given to the ongoing decrease in the pH of the Earth's oceans, caused by their uptake of anthropogenic carbon dioxide from the atmosphere. Although the natural absorption of CO2 by the world's oceans helps mitigate the climatic effects of anthropogenic emissions of CO2, it is believed that the resulting decrease in pH, (i.e. making the water acidic), will have negative consequences, primarily for oceanic calcifying organisms such as coral reefs.

What other causes are damaging coral reefs?

Coral Disease: During the last 10 years, the frequency of coral disease appears to have increased dramatically, contributing to the deterioration of coral reef communities around the globe. Most diseases occur in response to the onset of bacteria, fungi, and viruses. However, natural events

and human-caused activities may exacerbate reef-forming corals' susceptibility to waterborne pathogens.

More information is needed to identify the mechanisms by which most diseases kill their hosts, and how they are transmitted. The onset of coral disease has been shown to spread following coral bleaching events, so the evidence of a connection between warmer-than-normal water and coral disease is growing stronger. There is also evidence to indicate that low water quality increases incidence. It is critical that governments and managers continue their efforts to reduce (or stop) the effects of other major reef threats (sediments, pesticides, nutrients, overfishing, etc.) while this scientific information is gathered, if we are to give coral reefs a fighting chance of survival.

Crown of Thorns Starfish (COTs): The Crown of Thorns Starfish is a voracious coral reef predator. Populations of the COTs have increased since the 1970s and large outbreaks of starfish can occur wiping out huge tracks of coral reef. Few animals in the sea are willing to attack the spiny and toxic crown-of-thorns starfish, but some shrimp, worms and species of reef fish do feed on larvae or small adults. The decline of these predators, through over-harvesting and pollution, is one factor contributing to the rise in the population of the starfish.

Alien invasive species: Species that, as a result of human activity, have been moved, intentionally or unintentionally, into areas where they do not occur naturally are called "introduced species" or "alien species". In some cases where natural controls such as predators or parasites of an introduced species are lacking, the species may multiply rapidly, taking over its new environment, often drastically altering the ecosystem and out-competing local organisms. The damage caused by invasive species can be devastating, through alteration of ecosystem dynamics, biodiversity loss, reduction of the resilience of ecosystems, and loss of resources, with environmental, economic as well as socio-cultural impacts.